NUHA Software Requirements Specification

Oct 24, 2026 · @Azzam

1. Introduction

This SRS states what the NUHA Android app shall do, how each requirement is verified, and which requirements the code already meets as of commit `e29b1a6` (24 October 2026). It follows ISO/IEC/IEEE 29148:2018.

1.1 Purpose

This is the reference for building and accepting NUHA. Developers, designers, the Arabic early-years and data-protection reviewers, and the project owner use it to agree scope, plan work, and test releases. Each requirement has an ID, a priority, acceptance criteria, a source, and a status.

1.2 Scope

NUHA (نُهى) is an Arabic-only Android app for a smart educational table with a touchscreen and stylus, for children with autism, ADHD and learning difficulties, used at home, in resource rooms and in rehabilitation centres [PLAN §Limits; README]. It replaces the Flutter app «لوحتي الذكية», which is a reference only [README; LEG §1].

In scope:

- Child-first onboarding, child profiles, observation questions and a suggested starting path.
- A home adventure path with nine screen-and-stylus activities [PLAN §Method; README].
- Rewards, an adult area behind a gate, understandable reports, comfort settings, audio, and local storage with optional later sync.

Out of scope:

- Table hardware: enclosure, ambient LED lighting, storage compartments [PLAN §Tools]. The software only assumes a touchscreen, stylus and speakers.
- Diagnosis or treatment. NUHA is educational support, not a medical device [README; RB §1].
- iOS and web versions, which may follow in `ios/` and `web/` [README].

- The legacy extras «فراشة التركيز» and «تحدّي التركيز», unless separately approved [README; RB §8].
- Leaderboards, lives, streak penalties and shops [UI §Rewards].
- Sending child data to generative AI [RB §12].

1.3 Definitions

Term	Meaning
NUHA, «نُهى»	This product. «نُهى» is also the guide's working name on screen; the final guide name is open (OI-03).
«لوحتي الذكية»	The legacy Flutter app, v2.4.0, package `com.smartchild.app` [LEG §2].
Child mode	Screens the child uses: onboarding welcome, home path, activities, rewards.
Adult area	Screens for a parent, teacher or therapist: profiles, reports, settings. Reached through the gate.
Gate, «بوابة ولي الأمر»	A check that keeps children out of the adult area. The legacy gate is an addition question [LEG §5].
Guide	The lead mascot, a five-petal flower with a face and no limbs [UI §Mascot system; CODE `NuhaGuide`].
Shape buddy	One of six avatar mascots a child picks: squircle, dome, clover, sun, bean, triangle [CODE `AvatarKinds`].
Observation questions	Ten yes / no / not-sure questions a grown-up answers about the child's usual behaviour [LEG §4].
Starting path	The skill area the home path begins with, suggested by the observation answers: communication, attention, early learning, or mixed [CODE `StartPath`].
Home path (saga map)	Child home: a winding road with one stop per activity [UI §Home].

Term	Meaning
Activity	One of the nine learning tasks in §3.1.5.
Comfort modes	The three appearance settings: colourful (default), muted, high contrast [UI §Palette].
Chunky button	A pill-shaped button with a darker bottom lip that sinks when pressed and fires on touch-down [UI §Buttons].
Hue roles	Each palette hue has a fill, lip, text-on-fill, deep and tint colour [UI §Palette].
Token board	A strip of 3–5 star slots on a task screen [UI §Rewards].
Prompt ladder	Help that steps up after an error and fades as the child succeeds [UI §Carried forward].
Reduced motion	The setting that removes non-essential animation. On by default in the current code [CODE `AppPreferences`].
dp, sp	Android density-independent pixels; sp also scales with the system font size.
The table	The target screen: 1280×800 dp landscape, as on the Pixel Tablet emulator profile [EXP §Setup].
PDPL	Saudi Personal Data Protection Law and its Implementing Regulation [RB §12, S17].
TalkBack	Android's screen reader.

1.4 References

Source tags in brackets cite these documents. Section marks (§) point to headings within them.

Tag	Document	Used for
[PLAN]	`docs/project-plan/original-project-plan.pdf`, first project-plan report, Digital Technical College for Girls, 1448 H / 2026	Goals, scope, the nine activities, ages 4–10, hardware concept
[README]	`README.md`	Status, design principles, open decisions
[RB]	`docs/research/research-brief.md`, 20 Oct 2026, with sources S1–S20	Children's needs, accessibility, measurement, privacy, Kotlin findings
[UI]	`docs/research/ui-style.md`, Playful Soft-Flat	Palette, type, shape, buttons, mascots, map, rewards, motion, sound
[LEG]	`docs/legacy-app-reference.md`, 20 Oct 2026	Legacy modes, profiles, questionnaire, ten games, reports
[EXP]	`docs/experiments.md`, 20 Oct 2026	Stylus, accessible selection, layout and offline experiments
[CODE]	`android/` at commit `e29b1a6`	What is built
[OWN]	Owner decisions in working sessions, 24 Oct 2026	Child-first onboarding, three onboarding screens, gate removed for now, Baloo Bhaijaan 2 bundled
[ISO]	ISO/IEC/IEEE 29148:2018, Requirements engineering	Structure and requirement quality
[WCAG]	W3C WCAG 2.2 [RB S6]	Contrast, motion, input criteria

1.5 Conventions

- **IDs:** `FR-<area>-nnn` for functional, `IR-<area>-nnn` for interfaces, `NFR-<area>-nnn` for non-functional requirements.
- **Priority:** Must (needed for the first release with children), Should (expected, can slip a release), Could (desirable).
- **Status:** Built (met by [CODE] and checked on the emulator), In progress (partly built or uncommitted), Planned.

- **Verification:** T test, D demonstration, I inspection, A analysis (§4).
- "Shall" marks a requirement. Rationale and notes are kept outside the requirement text. Open decisions are listed in Appendix B with IDs `OI-nn` and are never decided here.

1.6 Overview

Section 2 describes the product, its users and its environment. Section 3 lists the requirements. Section 4 says how each is verified. Appendix A traces every requirement to its source and status; Appendix B lists open issues.

2. Overall description

NUHA is a standalone, offline-first Android app that runs on the table's touchscreen; a child uses it with stylus or finger while a grown-up sets it up and reads reports.

2.1 Product perspective

- **Replaces** «لوحتي الذكية» (Flutter, cloud sign-in, Firestore sync, optional Gemini analysis) [LEG §3–4]. Legacy features are inputs to evaluate, not requirements [README; LEG §1].
- **Runs on** the table's Android system. It depends on the touchscreen, a stylus, speakers, and the device's text-to-speech engine until recorded voice exists [PLAN §Tools; CODE `Speech`].
- **Stores data locally** in Room (`nuha-app.db`). A sync backend is optional and undecided (OI-17) [RB §11; EXP §4].
- **Built so far:** the three onboarding screens, the identity system, preferences and child profiles. The home path, activities, adult area, reports and sync are not built [CODE].

2.2 Product functions

1. Welcome the child with the guide, then set up who is playing and suggest a starting path.
2. Show a home path of activity stops; run each activity with one instruction, large targets and help.
3. Give feedback and rewards that never shame mistakes.
4. Let adults, behind a gate, manage profiles, read reports and change comfort settings.
5. Keep working with no network and never lose or duplicate a result.

2.3 User classes

User class	Characteristics	Main tasks	Access
Child	Age range open (OI-01). May not read yet; varies in receptive language, familiarity with standard Arabic, motor control, sensory tolerance and help needed [RB §3].	Meet the guide, pick a buddy, play activities, earn stars, pause or stop	Child mode only; can always pause [UI §Carried forward]
Parent or guardian	Sets up the device and profiles; may have limited time or reading confidence [RB §3].	Enter name and age, answer observation questions, read reports, adjust comfort	Adult area through the gate
Teacher or therapist	Works with one child at a time in a resource room or centre; chooses goals and records help given [RB §3; UI §Rewards].	Choose the sensory starting point, lock the path to one stop, read reports	Adult area through the gate

Which of these owns a profile on a shared table depends on the operator, which is undecided (OI-08) [RB §12].

2.4 Operating environment

Item	Value	Source
OS	Android 8.0 (API 26) minimum; target API 36; compile API 37	[CODE `build.gradle.kts`; EXP §Setup]
Orientation	Landscape only (`sensorLandscape`)	[CODE manifest]
Reference screen	1280×800 dp (2560×1600 px at density 320), Pixel Tablet profile	[EXP §Setup]
Input	Stylus and finger; stylus-only is undecided (OI-06)	[PLAN; README]
Audio	Built-in or table speakers	[PLAN §Tools]

Item	Value	Source
Network	Not required for any child-mode feature	[RB §11]
Table hardware	Not selected (OI-07)	[README; RB §10]

2.5 Constraints

- **Language and direction:** Arabic only, right-to-left on every screen regardless of device locale [README; CODE `NuhaTheme`].
- **Technology:** Kotlin, Jetpack Compose with Material 3, Room for storage, `androidx.ink` for drawing [README; EXP §Setup].
- **Child privacy:** child data stays on the device unless guardian consent, purpose and data handling are defined under the PDPL [README; RB §12].
- **Not a diagnosis:** no screen, report or result may present itself as diagnosis or treatment [README; RB §1, §5].
- **Content review:** Arabic letters, words, recordings, diacritics and numerals need approval by an Arabic early-years educator before release [README; RB §6].
- **Artwork:** original vector art drawn in Compose; no third-party illustrations or emoji [CODE `Identity.kt`; UI §What the style is].
- **Fonts:** bundled under the SIL Open Font License with the licence shipped in `assets/licenses/` [CODE].

2.6 Assumptions and dependencies

- The table's Android device supports the Ink API, landscape layout at about 1280×800 dp, and audio output. Unverified until hardware is chosen (OI-07).
- An Arabic text-to-speech voice is installed. On the emulator, Google TTS speaks with the `ar-xa-x-arz` voice; this is a stand-in, and the final register is open (OI-04) [CODE `Speech`].
- Emulator results establish behaviour, not stylus latency, palm rejection, reach or TalkBack speech [EXP §Limits].
- The Ink 1.0 renderer crashed once on first launch (`Unbalanced saves and restores`); drawing activities depend on it staying stable on the real device [EXP §1].

- Educators, a data-protection reviewer and families are available for the reviews and field tests the research proposes [RB §14].

3. Specific requirements

3.1 Functional requirements

Onboarding, profiles and the observation questions are built; the remaining features in 3.1.4–3.1.11 are planned. Status and verification for every ID are in Appendix A.

3.1.1 Onboarding (FR-ONB)

ID	Requirement	Pri	Acceptance criteria	Source
FR-ONB-001	On first launch the app shall run onboarding as three steps in order: welcome, who's playing, observation questions with starting path.	Must	Fresh install opens step 1. Step dots show 3 positions, the current one as a wide pill, read right to left.	[OWN; CODE]
FR-ONB-002	The app shall not show any onboarding screen until saved preferences have loaded.	Must	With `muted` saved, the first rendered welcome frame uses the muted palette.	[UI §Conflicts; CODE]
FR-ONB-003	The welcome screen shall show the guide at the start of a road, a bubble reading «أهلاً! أنا نُهى», three road stops with activity pictograms, and one primary button «هيّا نبدأ».	Must	Screen inspection on the reference screen; no other control can be activated.	[OWN; CODE]

ID	Requirement	Pri	Acceptance criteria	Source
FR-ONB-004	When voice is on, the welcome screen shall speak the greeting on entry and each time the guide is tapped.	Should	Voice on: a speech request within 1 s of entry and on each guide tap. Voice off: none.	[UI §Sound; CODE]
FR-ONB-005	With reduced motion off, the welcome screen shall: pop the guide in; grow the bubble from its tail after 250 ms; raise the button after 450 ms; after 5 s without input, turn the guide's gaze to the button and bob the button once; on start, hop the guide along the road to the first stop in about 900 ms before step 2.	Should	Screen recording shows each event at the stated time ±100 ms.	[UI §Motion; CODE]
FR-ONB-006	With reduced motion on, onboarding shall show all elements in their final state and start step 2 immediately when the button is pressed.	Must	Screen recording shows no entrance, idle or start animation.	[UI §Motion; CODE]
FR-ONB-007	The primary button shall sit in the same place on all three steps: horizontally centred at 531/640 of the screen width, bottom-aligned 24 dp above the navigation bar.	Must	Screenshots of the three steps: button bounds differ by ⩽ 4 dp.	[UI §Carried forward (targets never move); CODE]

ID	Requirement	Pri	Acceptance criteria	Source
FR-ONB-008	Steps 2 and 3 shall show a back control at the bottom left with an arrow pointing right, and system Back shall do the same: step 2 returns to step 1; step 3 returns to the previous question, then to step 2.	Must	Test each Back path; entered data is kept.	[UI §Carried forward (mirror sequence); CODE]
FR-ONB-009	Steps 2 and 3 shall show the guide in the top corner with a prompt bubble: «مَن سيلعب معي؟» (step 2), «أسئلة للكبار» (questions), «وجدنا البداية!» (result).	Must	Screen inspection.	[CODE]
FR-ONB-010	Finishing step 3 shall save the child profile and mark onboarding complete; later launches shall open child home, not onboarding.	Must	Finish, force-stop, relaunch: home opens. Today home is the placeholder «الخريطة قيد البناء».	[CODE]
FR-ONB-011	Data entered in steps 2 and 3 shall survive rotation and Back navigation within a session.	Should	Rotate and navigate back and forth: name, age, buddy and answers unchanged.	[CODE]
FR-ONB-012	An adult shall be able to re-run onboarding from the adult area without deleting existing profiles.	Should	From the adult area, start onboarding; existing profiles remain.	[LEG §4 (Managing profiles)]

Rationale: onboarding is child-first, with setup after the welcome [OWN]. The fixed button position applies the rule that targets never move [EXP §2; UI §Carried forward].

3.1.2 Child profiles (FR-PRF)

ID	Requirement	Pri	Acceptance criteria	Source
FR-PRF-001	Step 2 shall collect a name (required, 1–20 characters, trimmed), an age, and an avatar chosen from six shape buddies.	Must	Input of 21 characters keeps 20; a blank name is not accepted.	[LEG §4; CODE]
FR-PRF-002	Age shall be chosen from round buttons for 2, 3, 4, 5 and 6, with 4 preselected.	Must	Five age buttons; 4 selected on entry. Range subject to OI-01, numerals to OI-02.	[LEG §4; CODE]
FR-PRF-003	«التالي» on step 2 shall be inactive until the name is non-blank and an avatar is chosen.	Must	Test all four combinations.	[CODE]
FR-PRF-004	The name field shall stay fully visible while the on-screen keyboard is open on the reference screen.	Must	With the soft keyboard shown, the field's bounds lie above the keyboard.	[CODE]
FR-PRF-005	Avatars shall be shape characters without gender, dress or skin-tone cues, each in one fixed hue.	Must	Inspection of the six avatars in all three comfort modes.	[UI §Mascot system; CODE `NuhaAvatar`]

ID	Requirement	Pri	Acceptance criteria	Source
FR-PRF-006	A selected age, avatar or answer shall be shown by three cues together: tint face, thicker lip, and a check badge in the right-hand corner.	Must	Inspection in all comfort modes; badge visible on white faces in high contrast.	[UI §Buttons; WCAG 1.4.1; CODE `ChoiceTile`]
FR-PRF-007	A profile shall store only: name, avatar, age, observation answers, starting path and creation time. It shall not store diagnosis labels.	Must	Schema inspection of `child_profiles`.	[RB §12; CODE `ChildProfile`]
FR-PRF-008	The app shall support several child profiles on one device and ask which child is playing before child mode.	Should	Create two profiles; child mode opens for the chosen one; records stay separate.	[LEG §4; RB §3]
FR-PRF-009	An adult shall be able to edit a profile's name, age and avatar, and delete a profile with all its records after a confirmation.	Must	Edit persists after restart; delete removes the profile and its activity records from the database.	[LEG §4; RB §12]

Rationale: the legacy form collected avatar, name and age 2–6 with default 4 [LEG §4]. Shape buddies avoid dress and gender questions for Saudi families [UI §Mascot system].

3.1.3 Observation questions and starting path (FR-OBS)

ID	Requirement	Pri	Acceptance criteria	Source
FR-OBS-001	Step 3 shall present ten observation questions one at a time, each answered with «نعم», «لا» or «غير متأكد», with a counter «السؤال n من 10» and a ten-segment progress strip.	Must	Ten question screens; counter and strip match the index.	[LEG §4; CODE]
FR-OBS-002	Each question shall ask about the child's usual behaviour, worded so that «نعم» always means more difficulty.	Must	Review of all ten texts by an Arabic early-years specialist (OI-05).	[LEG §4; CODE `Questions`]
FR-OBS-003	Choosing an answer shall advance to the next question after 350 ms; «التالي» shall advance from an already answered question.	Should	Test both paths; earlier answers are kept on return.	[CODE]
FR-OBS-004	When voice is on, each question shall be read aloud when it appears.	Should	Speech request per question shown.	[RB §4; CODE]
FR-OBS-005	Every question screen shall show: «أجب بحسب ما تلاحظه عادةً على طفلك. تساعدنا الإجابات في اختيار نقطة البداية، ولا تشخّص أي حالة.»	Must	Text present on all ten screens.	[README; RB §5; CODE]

ID	Requirement	Pri	Acceptance criteria	Source
FR-OBS-006	The starting path shall be computed as follows. Each area's score is the sum of answer weights («نعم» 2, «غير متأكد» 1, «لا» 0) divided by twice its question count. Questions 1, 2, 3, 6, 7 are communication; 4, 5, 9, 10 attention; 8 early learning. The top area wins if its score is ⩾ 0.5 and ⩾ 0.15 above the next; otherwise the path is mixed.	Must	Unit tests: all «لا» → mixed; yes on 1, 2, 3, 6, 7 only → communication; yes on 8 only → early learning. Thresholds subject to OI-05.	[CODE `startPathFor`]
FR-OBS-007	The result shall show «تبدأ مع {name} رحلة», the path label («التواصل والمشاعر», «الانتباه والحركة», «التعلّم المبكر» or «قليل من كل شيء»), a one-line reason, the path tile, the child's avatar, the note «هذه نقطة بداية مقترحة فقط، ويمكن تغييرها لاحقًا.» and the button «لنبدأ الرحلة!».	Must	Inspection for each of the four paths; mixed shows all three area tiles.	[CODE]
FR-OBS-008	The starting path shall only choose where the home path begins and what is highlighted; it shall not hide or lock any activity.	Must	With each path, all activities remain reachable from home.	[RB §7 (shared rule); README]

ID	Requirement	Pri	Acceptance criteria	Source
FR-OBS-009	An adult shall be able to change a child's starting path from the adult area.	Should	Change persists and reorders home.	[CODE (result-screen promise)]
FR-OBS-010	If saving the profile fails, the result screen shall show «تعذّر الحفظ. حاول مرة أخرى.» and allow retry.	Should	Simulated write failure shows the message; retry saves once.	[CODE]

Rationale: the ten questions come from the legacy form, which asked about usual behaviour rather than a condition [LEG §4]. The research treats such answers as potentially sensitive and warns against diagnosis-based gating [RB §7, §12].

3.1.4 Home path (FR-HOM)

ID	Requirement	Pri	Acceptance criteria	Source
FR-HOM-001	Child home shall be a saga map: a winding road from bottom right to top left with one circular stop per activity, each in its activity hue with its pictogram, its mascot and up to three stars.	Must	Inspection on the reference screen.	[UI §Home]
FR-HOM-002	Stops shall keep fixed positions across sessions; new content shall appear as new stops, never by rearranging existing ones.	Must	Positions identical before and after progress and app updates.	[UI §Home; EXP §2]

ID	Requirement	Pri	Acceptance criteria	Source
FR-HOM-003	All nine stops shall fit on the 1280×800 dp screen without scrolling.	Must	Screenshot at font scale 1.0 and 1.5; no scroll container.	[UI §Home]
FR-HOM-004	Child navigation shall be two levels deep: map, then activity.	Must	Every activity opens from the map in one tap and returns to it in one action.	[UI §Home]
FR-HOM-005	The map shall begin at, and highlight, the stop for the child's starting path; mixed highlights the first stop.	Must	For each path, the highlighted stop matches FR-OBS-006.	[CODE `StartPath`; UI §Home]
FR-HOM-006	An adult shall be able to lock the map to one stop so the child sees only that activity.	Should	With a lock set, only that stop is shown and active.	[UI §Home]
FR-HOM-007	The map shall show the child's avatar and name.	Could	Inspection.	[LEG §5]
FR-HOM-008	The landscape art shall appear only on the map and celebration screens, never on task screens.	Must	Inspection of every task screen.	[UI §Conflicts]

3.1.5 Activities (FR-ACT and per activity)

The nine activities come from the original plan [PLAN §Method; README]. Their final learning objectives are open (OI-10), so each per-activity requirement states only the core interaction and what to record.

Common to all activities

ID	Requirement	Pri	Acceptance criteria	Source
FR-ACT-001	Before content production, each activity shall have an approved sheet: one learning objective, prerequisites, successful behaviour, likely confounds, examples, accessibility alternative, and a transfer question.	Must	Sheet signed off by the Arabic or maths educator.	[RB §8]
FR-ACT-002	A task screen shall give one instruction at a time, with a demonstration and a spoken version the child can repeat.	Must	Each task step has a replay control for both.	[RB §4; UI §Carried forward]
FR-ACT-003	The task, answer targets, help and pause shall be visible together without scrolling.	Must	Screenshots at font scale 1.0 and 1.5 on the reference screen.	[RB §4; EXP §3; LEG §9]
FR-ACT-004	Targets shall not move during a task; a placed item shall leave an empty slot and remaining items shall not reflow.	Must	Place each item in turn; others keep their bounds.	[EXP §2; UI §Buttons]
FR-ACT-005	Where dragging or drawing is not the skill being taught, the activity shall also accept tap-item then tap-destination.	Must	Complete each such activity using taps only.	[RB §4; EXP §2; UI §Buttons]

ID	Requirement	Pri	Acceptance criteria	Source
FR-ACT-006	Taps shall fire on touch-down; child mode shall not require double-tap, long-press or pinch.	Must	Gesture audit of every child screen.	[UI §Carried forward]
FR-ACT-007	A wrong answer shall return gently to its slot, step up the prompt ladder and be recorded as "correct with help" once solved; there shall be no failure state or buzzer.	Must	Wrong-answer test per activity.	[UI §Rewards]
FR-ACT-008	Feedback shall begin within 0.5 s of a valid response.	Must	Timed screen recording.	[UI §Carried forward]
FR-ACT-009	A pause control shall be visible at all times and work for the child; stopping shall keep earned stars and records.	Must	Pause and quit mid-activity; records and stars persist.	[RB §4, §9; UI §Carried forward]
FR-ACT-010	If the app is interrupted mid-activity, completed trials shall be kept and the activity shall offer to resume.	Must	Force-stop mid-activity, relaunch: completed trials present.	[RB §4]

ID	Requirement	Pri	Acceptance criteria	Source
FR-ACT-011	Each trial shall be recorded with: an ID created on the device when the trial starts, task version, difficulty settings, input type (stylus or finger), first independent choice, hints and help given, completion, and active time excluding pauses.	Must	Database inspection after a scripted session.	[RB §8–9; EXP §4]
FR-ACT-012	Difficulty shall change one factor at a time, and the active settings shall be stored with each trial.	Should	Records show one changed factor between consecutive levels.	[RB §8]
FR-ACT-013	A task screen shall show only one mascot, which speaks only at instruction and feedback moments and stays still while the child works.	Must	Inspection and screen recording.	[UI §Mascot system, §Conflicts]
FR-ACT-014	Task screens shall use the paper background with only the task objects and one mascot.	Must	Inspection.	[UI §Shape and elevation]
FR-ACT-015	Learning content (letters, words, sentences being learned) shall use a simplified Naskh face, fully vowelised, at 28–32 sp with 1.6–1.8× line height; the rounded face is for the interface only.	Must	Typography inspection of every learning screen.	[UI §Typography]

ID	Requirement	Pri	Acceptance criteria	Source
FR-ACT-016	An adult shall be able to replace any activity mascot with the lead guide.	Should	Setting changes the mascot on all task screens.	[UI §Conflicts]
FR-ACT-017	Drawing activities shall use the Ink API, save strokes as normalised 0–1 points with time and tool type, support undo and clear, and replay a stroke from saved data.	Must	Replayed stroke matches the original after a canvas resize.	[EXP §1; RB §11]
FR-ACT-018	Drawing checks shall accept reasonable variation in a stroke rather than require exact pixel overlap.	Should	Educator-drawn sample strokes within tolerance are accepted.	[RB §6]

Per activity

ID	Activity	Requirement	Pri	Acceptance criteria	Source
FR-EMO-001	«تلوين المشاعر»	The child shall connect or select the feeling that matches an illustrated face from at least three choices, with characters and contexts varied across trials.	Must	Session uses ≥ 2 characters per emotion.	[PLAN; LEG §6.2; RB §7]
FR-EMO-002	«تلوين المشاعر»	The activity shall not teach a colour as the universal meaning of an emotion, and shall record first choice, hints, revised choice and examples used.	Must	Content review; record inspection.	[RB §7]

ID	Activity	Requirement	Pri	Acceptance criteria	Source
FR-ENV-001	«رحلة مطابقة البيئة»	The child shall place items into environments (for example rain, sun, garden) by drawing, dragging or tap-tap; a question shall either name one relationship or accept every justified match.	Must	Each item has a defined accepted set.	[PLAN; LEG §6.3; RB §7; EXP §2]
FR-ENV-002	«رحلة مطابقة البيئة»	Conceptual choices shall be recorded separately from missed drop targets.	Must	Record inspection after a missed drop.	[RB §7]
FR-PTH-001	«تتبع مسار الأشكال الاجتماعية»	The child shall draw a route between two characters while avoiding obstacles; start and destination shall be clear and an example shown first; route complexity and corridor width shall vary separately.	Must	Level settings change one of the two at a time.	[PLAN; LEG §6.4; RB §7]
FR-PTH-002	«تتبع مسار الأشكال الاجتماعية»	Records shall hold route, boundary contacts, restarts, help and input type; reports shall describe it as route planning and motor practice, not a social-skills measure.	Must	Record and report inspection.	[RB §7]
FR-STR-001	«تتبع النجوم الهادئة»	The child shall keep the pen inside a circle that follows a slowly moving star; target size, speed and path shall be set independently, with a demonstration and rests.	Must	Settings expose the three factors separately.	[PLAN; LEG §6.5; RB §8]
FR-STR-002	«تتبع النجوم الهادئة»	The record shall hold time in target while contact is active, interruptions, speed and target size; no report shall call it an attention measure.	Must	Record and report inspection.	[RB §5, §8]

ID	Activity	Requirement	Pri	Acceptance criteria	Source
FR-MAZ-001	«متاهة توازن الكرة»	The child shall draw lines to guide a ball past holes, then run the ball in a predictable, replayable simulation. Whether lines are a route or walls is open (OI-10).	Must	Same drawing gives the same result on replay.	[PLAN; LEG §6.6; RB §8]
FR-MAZ-002	«متاهة توازن الكرة»	Records shall separate the drawing (valid route) from the simulation outcome (collisions), with retries.	Must	Record inspection.	[RB §8]
FR-GDN-001	«زرع بستان الأزهار»	The child shall select points or numbers in order to grow flowers, with all targets on screen, no speed requirement by default, and timing only in a separately chosen mode.	Must	Default mode has no timer; all targets visible without scrolling.	[PLAN; LEG §6.7; RB §8; EXP §3]
FR-GDN-002	«زرع بستان الأزهار»	Records shall hold correct sequence choices, skipped items and help; timing only in the timed mode.	Must	Record inspection.	[RB §8]
FR-LET-001	«كتابة الحروف وتوصيل الكلمات»	Letter formation, letter recognition, missing-letter completion and word–picture matching shall be separate task types.	Must	Each type selectable and recorded separately.	[PLAN; LEG §6.8; RB §6, §8]
FR-LET-002	«كتابة الحروف وتوصيل الكلمات»	Tracing guides shall be educator-approved vector letter forms with stroke direction, not text drawn in a system font; isolated and connected forms shall follow the curriculum.	Must	Asset inspection; educator sign-off.	[EXP §1; RB §6]
FR-PAT-001	«نسخ الأنماط الهندسية»	Pattern continuation (choose the next shape) and shape drawing shall be separate modes, and a level shall vary one pattern rule at a time.	Must	Mode switch; level definitions.	[PLAN; LEG §6.9; RB §8]

ID	Activity	Requirement	Pri	Acceptance criteria	Source
FR-BAL-001	«الميزان الحسابي البصري»	The balance shall show equal quantities on both pans when solved, let the child inspect the groups before answering, and accept the number by selection or by writing; handwriting errors shall be recorded separately from maths errors.	Must	Both input modes; record separates the two error kinds.	[PLAN; LEG §6.10; RB §6, §8]

3.1.6 Rewards (FR-RWD)

ID	Requirement	Pri	Acceptance criteria	Source
FR-RWD-001	A task screen shall show a token board of 3–5 star slots; each correct or helped-correct trial shall fill one slot, and a full board shall trigger a celebration and a break.	Must	Scripted session fills the board and reaches the break.	[UI §Rewards]
FR-RWD-002	A per-trial celebration shall last ⩽ 1.5 s: a star flies into its slot and the mascot nods, with no confetti; under reduced motion the star fades in.	Must	Timed recording in both motion modes.	[UI §Rewards, §Motion]
FR-RWD-003	A per-session celebration shall show a short confetti burst, a ribbon «أحسنت!» and the stars earned; under reduced motion the ribbon is static with no confetti.	Must	Recording in both motion modes.	[UI §Rewards, §Motion]

ID	Requirement	Pri	Acceptance criteria	Source
FR-RWD-004	Milestone badges shall be kept in a trophy case on the adult-visible profile, and each badge shall have one name used on every screen.	Should	Badge names identical in child and adult views.	[UI §Rewards; LEG §8]
FR-RWD-005	Celebration intensity shall be adjustable (off, gentle, lively), default gentle.	Must	Each setting changes FR-RWD-002/003 output.	[UI §Conflicts]
FR-RWD-006	Praise shall state what was right.	Must	Content review of feedback lines.	[UI §Rewards; RB §9]
FR-RWD-007	The app shall have no leaderboards, lives, streak penalties, shops or comparison between children.	Must	Inspection of all screens.	[UI §Rewards; README; RB §9]
FR-RWD-008	Stars on the map shall reflect only the child's own progress, up to three per activity.	Must	Two profiles keep separate stars.	[UI §Home, §Rewards]

3.1.7 Adult area and gate (FR-ADM)

A corner gate (3-second hold) was built on the welcome screen and removed on 24 October 2026 at the owner's request; there is no adult entry point today [OWN].

ID	Requirement	Pri	Acceptance criteria	Source
FR-ADM-001	Profiles, reports and settings shall be reachable only through the gate, in an adult area styled as a separate Material 3 theme.	Must	No child-mode route reaches these screens without the gate.	[UI §Carried forward]
FR-ADM-002	The gate shall require a deliberate adult action that random tapping, dragging or holding cannot complete. The mechanism is open (OI-15).	Must	10,000 random input events in child mode never open the adult area; 5 adults pass it in ⩽ 10 s each.	[LEG §5; UI §Carried forward]
FR-ADM-003	The gate shall not be described or used as guardianship verification or consent.	Must	Copy review of the gate and consent screens.	[RB §12]
FR-ADM-004	The child shall be able to pause or stop play without passing the gate.	Must	Pause and exit work from every child screen with no gate.	[RB §4; UI §Carried forward]
FR-ADM-005	The gate shall offer an accessible path for TalkBack users.	Should	Gate passable with TalkBack on.	[RB §4]
FR-ADM-006	The adult area shall return to child mode for a chosen child in one action.	Should	One tap from adult home to child home.	[LEG §4 (Navigation and dashboard)]

3.1.8 Reports (FR-REP)

ID	Requirement	Pri	Acceptance criteria	Source
FR-REP-001	Reports shall state counts with any percentage: items presented, first independent choices correct, correct with help, task version, difficulty and input type, for example «طابق 4 من 6 أمثلة جديدة بمساعدتين».	Must	Every figure in a report has its counts beside it.	[README; RB §9]
FR-REP-002	Reports shall not show composite scores without a stated formula (the legacy "64% overall progress" is not to be inherited).	Must	Each percentage links to its definition.	[RB §9; LEG §8]
FR-REP-003	Reports shall never rank or compare children.	Must	Inspection of all report views.	[README; RB §9]
FR-REP-004	Reports shall name the skill each activity practises and shall not describe results as a diagnosis, attention measure or developmental age.	Must	Copy review.	[RB §5, §7]
FR-REP-005	Reports shall list sessions with date, activity, active time and result counts.	Should	Scripted sessions appear with correct values.	[LEG §4 (Reports); RB §9]
FR-REP-006	Trends over the last 7 sessions and the last 4 weeks shall compare only trials with the same task version, difficulty and input type.	Should	Mixed-condition data is shown as separate series.	[LEG §4; RB §9]

ID	Requirement	Pri	Acceptance criteria	Source
FR-REP-007	An adult shall be able to copy a plain-text summary of a child's report.	Could	Copied text matches the on-screen counts.	[LEG §4 («نسخ الملخص»)]
FR-REP-008	Before release, a sample report shall be shown to adults to check they read it correctly.	Should	At least 4 adults explain the sample correctly (Gate C).	[RB §9, §14]

3.1.9 Settings and comfort modes (FR-SET)

ID	Requirement	Pri	Acceptance criteria	Source
FR-SET-001	The app shall offer three appearance modes: colourful (default, the palette table), muted (each fill moved 55% toward its tint, lips unchanged, ink text on fills), and high contrast (deep shades as fills with white text, ink lips, white paper).	Must	Palette tokens match `ui-style.md`; `tools/android/check_identity.py` passes.	[UI §Palette; CODE `paletteFor`]
FR-SET-002	An adult shall be able to set, per child: appearance, motion, voice, effects, music, celebration intensity, mascot on or off, prompt delay, target size and numerals.	Must	Each setting changes behaviour for that child only.	[UI §Carried forward]

ID	Requirement	Pri	Acceptance criteria	Source
FR-SET-003	Setting changes shall apply immediately, persist across restarts, and be saved in the order they were made.	Must	Rapid toggles then force-stop: last value restored.	[CODE `PreferencesViewModel`]
FR-SET-004	Motion shall have three levels (full, reduced, none), defaulting to the system animation setting.	Must	With system animations off, a new install starts at none. Conflicts with the current default (OI-14).	[UI §Motion]
FR-SET-005	Before a child's first session, the adult shall choose a sensory starting point (lively, gentle or muted).	Should	Setup cannot finish without a choice. Placement in the child-first flow is open (OI-14).	[UI §Conflicts]
FR-SET-006	Music shall default to off and voice to on.	Must	New install: music off, voice on.	[UI §Sound; CODE `AppPreferences`]

3.1.10 Audio (FR-AUD)

ID	Requirement	Pri	Acceptance criteria	Source
FR-AUD-001	Voice, effects and music shall be separate channels, each switchable on its own.	Must	Each switch silences only its channel.	[UI §Sound; RB §4; CODE]
FR-AUD-002	Effects shall be short and soft-attack with no buzzer; any sound under speech shall be at least 20 dB quieter than the speech, or off.	Must	Level measurement of mixed playback.	[UI §Sound]

ID	Requirement	Pri	Acceptance criteria	Source
FR-AUD-003	Released voice prompts shall be recorded Arabic audio approved by an Arabic early-years educator; device text-to-speech is a development stand-in only.	Must	Release build plays recorded files for every prompt. Register open (OI-04).	[README; RB §6; CODE `Speech`]
FR-AUD-004	Every spoken instruction shall also be understandable with all audio off, through text, demonstration or both.	Must	Muted-audio walkthrough of each activity.	[RB §4]
FR-AUD-005	All audio shall be packaged on the device and play with no network.	Must	Airplane-mode playback test.	[RB §11]
FR-AUD-006	Pressing a chunky button or choice shall play a soft pop when effects are on.	Should	Effects on: pop on press; off: silent.	[UI §Sound; CODE `Effects`]

3.1.11 Storage and sync (FR-SYN)

ID	Requirement	Pri	Acceptance criteria	Source
FR-SYN-001	Every child-mode feature shall work with no network connection.	Must	Full session in airplane mode.	[RB §11]
FR-SYN-002	Each trial result shall be written to local storage when the trial ends, before any upload.	Must	Force-stop right after a trial: the result is present.	[EXP §4]

ID	Requirement	Pri	Acceptance criteria	Source
FR-SYN-003	If sync is added, uploads shall be idempotent by the device-created trial ID, so a retry after a lost acknowledgement creates no duplicate.	Must	Interrupt after upload, before confirm, then resync: server count unchanged.	[EXP §4]
FR-SYN-004	Sync shall be off by default and enabled only after guardian consent that states what leaves the device, to whom, why and for how long.	Must	Fresh install sends no data; consent screen copy reviewed.	[RB §12; README]
FR-SYN-005	Sync across several devices shall follow a defined conflict rule.	Should	Two devices editing one profile converge per the rule (OI-17).	[EXP §4; LEG §4]

3.2 External interface requirements

3.2.1 User interface (IR-UI)

ID	Requirement	Pri	Acceptance criteria	Source
IR-UI-001	Every screen shall lay out right to left, whatever the device locale.	Must	Device set to English: layout still RTL.	[README; CODE `NuhaTheme`]
IR-UI-002	Interface text shall use one rounded Arabic face in heavy weights: Black for titles of 24 sp and above, ExtraBold for buttons and below 24 sp, SemiBold for secondary labels; no letter-spacing or tatweel.	Must	Typography inspection. Face choice open (OI-11); today Baloo Bhaijaan 2, 400–800, so Black renders at 800.	[UI §Typography; CODE `Type.kt`]

ID	Requirement	Pri	Acceptance criteria	Source
IR-UI-003	Interface labels shall be at least 20 sp; buttons in child mode at least 24 sp.	Must	Inspection of computed sizes.	[UI §Typography]
IR-UI-004	Cards and tiles shall have 24 dp corners, buttons full pill ends, stops and avatars circles; depth shall be one soft hue-tinted shadow and a top-to-bottom gradient of at most about 6% lightness, with no outlines except the lip edge.	Must	Measure corners and gradient. The onboarding card (40 dp) and tiles (28 dp) do not yet match.	[UI §Shape and elevation; CODE `ChunkySurface`]
IR-UI-005	Chunky buttons shall have a 4–6 dp lip; on touch-down the face sinks by the lip height and the action fires, without changing the hit area or moving neighbours; repeat fires within 300 ms shall be ignored.	Must	Recording and bounds check; double-tap fires once.	[UI §Buttons; CODE `ChunkyButton`]
IR-UI-006	Faces below 3:1 against paper shall carry a 2 dp edge in the lip colour.	Must	Contrast check of each face against paper.	[UI §Palette; WCAG 1.4.11; CODE `ChunkySurface`]
IR-UI-007	Each activity shall keep one hue on its tile, stop, badge and header; red and pink are activity colours, and there shall be no error colour.	Must	Hue audit per activity.	[UI §Palette]

ID	Requirement	Pri	Acceptance criteria	Source
IR-UI-008	Sequence elements (back and next, progress, the path) shall mirror for RTL; media controls, check marks, letter-tracing paths and charts shall not.	Must	Inspection: back arrow points right; check marks unmirrored.	[UI §Carried forward; CODE `BackCircle`]
IR-UI-009	Child-mode controls shall be inset from screen edges and corners.	Must	No child control within 24 dp of a screen edge.	[UI §Carried forward]
IR-UI-010	Artwork shall be flat vector drawn in code; illustrations shall use content coordinates that do not mirror with RTL.	Must	Code inspection of `Identity.kt`.	[UI §What the style is; CODE]
IR-UI-011	An inactive button shall show its tint face with faint ink, not transparency over the scene.	Should	Inspection on yellow and white backgrounds.	[CODE `ChunkyButton`]

3.2.2 Hardware, stylus and touch (IR-HW)

ID	Requirement	Pri	Acceptance criteria	Source
IR-HW-001	The app shall accept stylus and finger input and record the tool type for every stroke and trial.	Must	Records show stylus or finger correctly on the real device.	[EXP §1]

ID	Requirement	Pri	Acceptance criteria	Source
IR-HW-002	Palm rejection and any stylus-only restriction shall be tested on the selected table before drawing geometry is fixed.	Must	Physical-test matrix completed on the chosen hardware (OI-06, OI-07).	[EXP §1; RB §10]
IR-HW-003	Ink latency acceptance thresholds shall be agreed after supervised observation on the real table, not set in advance.	Must	Thresholds recorded in the test plan before release.	[RB §10, §14]
IR-HW-004	Layouts shall be checked for left- and right-handed reach and for the hand covering targets.	Should	Physical test with both hands.	[RB §4, §10]
IR-HW-005	Pressure or tilt shall be used only if the chosen hardware reports them reliably.	Could	Hardware evaluation report.	[RB §10]

3.2.3 Audio (IR-AUD)

ID	Requirement	Pri	Acceptance criteria	Source
IR-AUD-001	Effects shall play on the sonification stream and voice through the device speech engine or media playback, following the system volume.	Must	System volume changes the app's output.	[CODE `Audio.kt`]
IR-AUD-002	When no Arabic speech voice is available, the app shall stay silent rather than speak another language.	Must	Remove Arabic voice data: no speech, no crash.	[CODE `Speech`]

3.2.4 Storage (IR-STO)

ID	Requirement	Pri	Acceptance criteria	Source
IR-STO-001	Data shall be stored in a local Room database (`nuha-app.db`) holding `app_preferences` (one row) and `child_profiles`.	Must	Schema inspection.	[CODE `AppDatabase`]
IR-STO-002	Every schema change shall ship with a migration that keeps existing data.	Must	Upgrade test from the previous release; v1 to v2 passed on 24 Oct 2026.	[CODE; EXP §4 (not yet tested there)]
IR-STO-003	Activity trials shall be stored locally, keyed by child and by the device-created trial ID.	Must	Schema inspection; duplicate ID insert is rejected.	[EXP §4; RB §11]
IR-STO-004	The database shall live in app-private storage and be excluded from device backups.	Must	Manifest has `allowBackup="false"` ; no external-storage writes.	[RB §12; CODE manifest]

3.3 Non-functional requirements

No threshold below is invented: where the sources give no number (ink latency, session length), the requirement is to agree one after field observation.

3.3.1 Accessibility (NFR-ACC)

ID	Requirement	Pri	Acceptance criteria	Source
NFR-ACC-001	Text shall have contrast of at least 4.5:1 (3:1 for large text, 18.66 px bold or more); controls and meaningful graphics at least 3:1; in all three comfort modes.	Must	`tools/android/check_identity.py` passes for every token pair in every mode.	[WCAG 1.4.3, 1.4.11; UI §Palette; CODE]
NFR-ACC-002	Colour shall never be the only cue; shape, icon, border or position shall carry the same meaning.	Must	Greyscale screenshots keep every state distinguishable.	[WCAG 1.4.1; UI §Carried forward]
NFR-ACC-003	Every interactive control shall be at least 72 dp; primary task targets 96–120 dp; gaps between targets 16–24 dp; target sizes computed from the available task area.	Must	Bounds audit of every screen.	[UI §Buttons; EXP §3; RB §4]

ID	Requirement	Pri	Acceptance criteria	Source
NFR-ACC-004	Text that scales with font size shall not sit inside an area from which target sizes are computed.	Must	Layout inspection at font scale 1.5.	[EXP §3]
NFR-ACC-005	At font scale 1.5 on the reference screen, no Arabic text shall clip and no control shall leave the screen.	Must	Screenshots of every screen at 1.5. Welcome verified; steps 2 and 3 not yet.	[RB §4; EXP §3]
NFR-ACC-006	Reduced motion shall remove entrance, idle and celebration motion and replace confetti with a static ribbon; nothing shall flash more than 3 times per second.	Must	Recordings in each motion level.	[UI §Motion; WCAG 2.3.1]

ID	Requirement	Pri	Acceptance criteria	Source
NFR-ACC-007	On task screens, motion shall respond only to touch or point at the learning target; UI response motion shall last under about 400 ms; only the session confetti may exceed 1.5 s.	Must	Recording audit of task screens.	[UI §Motion]
NFR-ACC-008	Each interactive element, including those drawn on a canvas, shall be one accessibility node with an Arabic label, its state, and actions that complete the task.	Must	UI hierarchy dump shows one node per element with label and state.	[EXP §2; RB §11]
NFR-ACC-009	TalkBack reading order, labels and spoken output shall be tested on a real device.	Must	TalkBack walkthrough of onboarding and one activity per type.	[RB §4; EXP §2]
NFR-ACC-010	Focus shall be visible when navigating with a keyboard or switch access.	Should	Switch Access walkthrough.	[RB §4; EXP §2]

3.3.2 Performance (NFR-PRF)

ID	Requirement	Pri	Acceptance criteria	Source
NFR-PRF-001	Ink shall appear while the stylus moves and keep up while audio and animation play.	Must	Physical test against the threshold agreed under IR-HW-003.	[EXP §1; RB §10]
NFR-PRF-002	Saving preferences, profiles and trials shall not block the UI thread, and saved state shall keep the order of the inputs that produced it.	Must	Rapid input test: no dropped frames from writes; final state matches last input.	[CODE `PreferencesViewModel`; RB §11]

3.3.3 Offline behaviour (NFR-OFF)

ID	Requirement	Pri	Acceptance criteria	Source
NFR-OFF-001	Activities, audio, fonts and art shall be bundled in the app; first run shall need no download.	Must	First launch in airplane mode completes onboarding and an activity.	[RB §11; LEG §4 (Settings)]
NFR-OFF-002	The release shall not request network permission until sync is approved under FR-SYN-004.	Must	Manifest inspection: no `INTERNET` permission.	[RB §12; CODE manifest]

3.3.4 Privacy and data protection (NFR-PRV)

ID	Requirement	Pri	Acceptance criteria	Source
NFR-PRV-001	Before any backend is chosen, a data inventory shall list each stored field with its purpose; fields without a purpose shall be removed.	Must	Inventory document reviewed.	[RB §12]
NFR-PRV-002	No child data, including observation answers, shall be sent to generative AI or third-party analytics.	Must	Code and dependency audit; network traffic capture shows none.	[RB §12; LEG §4 (Gemini option)]
NFR-PRV-003	Observation answers shall be treated as potentially sensitive: kept on the device, shown only in the adult area, and deleted with the profile.	Must	Inspection; delete test.	[RB §12]
NFR-PRV-004	Access protection (the gate), research permission and cloud processing consent shall be separate, each explaining what it covers.	Must	Copy review of each screen.	[RB §12]
NFR-PRV-005	Deletion, withdrawal, export, shared-device access and retention shall be defined before any cloud launch.	Must	Policy document approved before sync ships.	[RB §12]
NFR-PRV-006	The app shall not write child data to logs.	Must	Log capture during a full session shows no names or answers.	[RB §12]

ID	Requirement	Pri	Acceptance criteria	Source
NFR-PRV-007	A qualified PDPL review, including guardian consent under Implementing Regulation Article 13, shall precede use with real families.	Must	Signed review on file.	[RB §12 (S17)]
NFR-PRV-008	Field sessions with children shall use guardian permission and child assent, accessible stop signals, no penalty for stopping, and separate permission for recordings.	Must	Ethics checklist per session.	[RB §12]

3.3.5 Safety: not a diagnosis (NFR-SAF)

ID	Requirement	Pri	Acceptance criteria	Source
NFR-SAF-001	No screen, report, store listing or document shipped with the app shall claim diagnosis, treatment or clinical benefit.	Must	Copy review of all text.	[README; RB §1, §5]
NFR-SAF-002	The app shall not derive a developmental age, health-like score or attention diagnosis from session data.	Must	Report and data-model inspection.	[RB §5, §8]
NFR-SAF-003	The child shall be able to stop at any time with no penalty, lost reward or forced completion.	Must	Stop mid-activity: no penalty message; rewards kept.	[RB §10, §12]

3.3.6 Localisation (NFR-LOC)

ID	Requirement	Pri	Acceptance criteria	Source
NFR-LOC-001	All interface text shall be Arabic; Latin script appears only in names an adult types.	Must	String audit.	[README]
NFR-LOC-002	One numeral system shall be used app-wide and never mixed on a screen.	Must	String and asset audit. Which system is open (OI-02); today Western digits.	[UI §Carried forward; RB §6]
NFR-LOC-003	Learning content shall carry full tashkeel checked by a native speaker; interface text light tashkeel.	Must	Content review sign-off.	[UI §Carried forward; RB §6]
NFR-LOC-004	Mixed text (Arabic with Latin names, numbers or symbols) shall render in correct bidirectional order.	Must	«تبدأ مع Sara رحلة» renders correctly; equations reviewed, not blindly mirrored.	[RB §6 (S11); CODE]

3.3.7 Maintainability (NFR-MNT)

ID	Requirement	Pri	Acceptance criteria	Source
NFR-MNT-001	The Android app shall be a self-contained Gradle project in `android/`, with shared docs and tools outside it.	Must	Repository inspection.	[README]
NFR-MNT-002	Palette, scene and comfort-mode tokens shall be defined in one file (`Identity.kt`) and checked by `check_identity.py`.	Must	Code inspection; script run.	[CODE]

ID	Requirement	Pri	Acceptance criteria	Source
NFR-MNT-003	Starting-path scoring, migrations and trial recording shall have automated unit tests.	Should	Test suite runs in CI. No tests exist today.	[ISO; CODE]
NFR-MNT-004	Arabic interface strings shall be kept in string resources so an educator can review them in one place.	Should	No user-facing Arabic literals in Kotlin. Today they are inline.	[README (content review)]
NFR-MNT-005	Library versions shall be pinned in `libs.versions.toml`, and the Ink renderer crash shall be rechecked on each new stable Ink release.	Must	Version catalogue inspection; crash log review.	[EXP §1, §Setup]
NFR-MNT-006	`docs/onboarding.md` and `tools/android/verify_onboarding.py` shall describe the current onboarding.	Should	Doc and script match the three built screens. Both are stale today.	[CODE]

4. Verification

Each requirement is closed by one of four methods in one of four environments; Appendix A gives the method for every ID, and a requirement is marked Built only after it passes on the emulator.

4.1 Methods

Method	Meaning	Typical evidence
T, Test	Scripted or automated check with defined inputs and expected outputs	Unit tests, `adb` scripts, database queries

Method	Meaning	Typical evidence
D, Demonstration	Run the app and observe the behaviour	Screen recordings, screenshots, reviewer notes
I, Inspection	Review code, assets, copy, schema or documents against the requirement	Review checklist, sign-off
A, Analysis	Measure or calculate	Contrast ratios, audio levels, timings taken from recordings

4.2 Environments

Env	Setup	Used for
E1 Emulator	Pixel Tablet profile, API 36, 2560×1600 at density 320 (1280×800 dp), landscape; input by `adb`; font scale 1.0 and 1.5; comfort modes set in the preferences table	Layout, flow, persistence, motion, contrast; all current Built statuses [EXP §Setup]
E2 Real table	The selected table and stylus, once chosen (OI-07)	Ink latency, palm rejection, reach, handedness, TalkBack speech, audio levels [EXP §Limits; RB §10]
E3 Field sessions	Supervised sessions with consent and assent, Gates B and C of the research plan	Comprehension, help needed, stopping, report interpretation [RB §14]
E4 Expert review	Arabic early-years educator, maths educator, specialist for the observation questions, PDPL reviewer	Content, question wording, scoring, privacy [RB §6, §12]

4.3 Rules

- Requirements needing E2, E3 or E4 cannot be closed on the emulator alone.
- Every release runs the regression set: onboarding end to end in all three comfort modes and both motion settings, font scale 1.5, a force-stop during each step, and `check_identity.py`.

- Acceptance thresholds the sources leave open (ink latency, session length, gate pass time for children) are recorded in the test plan once agreed, never assumed [RB §10, §14].

Appendix A. Traceability matrix

Of 169 requirements, 50 are built, 26 in progress and 93 planned as of commit `e29b1a6`; change the Status dropdown as work lands. Method letters follow §4.1; E2–E4 mark checks that need the real table, field sessions or expert review.

A.1 Onboarding, profiles, observation, home, common activity rules

ID	Title	Pri	Source	Method	Status
FR-ONB-001	Three-step onboarding	Must	OWN, CODE	D	Built
FR-ONB-002	Wait for saved preferences	Must	UI, CODE	T	Built
FR-ONB-003	Welcome screen content	Must	OWN, CODE	I	Built
FR-ONB-004	Spoken greeting	Should	UI, CODE	D	Built
FR-ONB-005	Welcome motion	Should	UI, CODE	A	Built
FR-ONB-006	Reduced-motion onboarding	Must	UI, CODE	D	Built
FR-ONB-007	Primary button never moves	Must	UI, CODE	A	Built
FR-ONB-008	Back control and system Back	Must	UI, CODE	T	Built
FR-ONB-009	Host guide prompts	Must	CODE	I	Built
FR-ONB-010	Finish saves and completes	Must	CODE	T	Built
FR-ONB-011	Entries survive rotation and Back	Should	CODE	T	Built
FR-ONB-012	Re-run onboarding from adult area	Should	LEG	D	Planned
FR-PRF-001	Name, age, avatar	Must	LEG, CODE	T	Built
FR-PRF-002	Age buttons 2–6, default 4	Must	LEG, CODE	I	Built
FR-PRF-003	Next needs name and avatar	Must	CODE	T	Built
FR-PRF-004	Name field above keyboard	Must	CODE	D	In progress

ID	Title	Pri	Source	Method	Status
FR-PRF-005	Shape-buddy avatars	Must	UI, CODE	I	Built
FR-PRF-006	Three selection cues	Must	UI, WCAG, CODE	I	Built
FR-PRF-007	Minimal profile fields	Must	RB, CODE	I	Built
FR-PRF-008	Several profiles per device	Should	LEG, RB	D	Planned
FR-PRF-009	Edit and delete a profile	Must	LEG, RB	T	Planned
FR-OBS-001	Ten questions, three answers	Must	LEG, CODE	D	Built
FR-OBS-002	Question wording	Must	LEG, CODE	I (E4)	In progress
FR-OBS-003	Auto-advance after 350 ms	Should	CODE	T	Built
FR-OBS-004	Questions read aloud	Should	RB, CODE	D	Built
FR-OBS-005	Not-a-diagnosis notice	Must	README, RB, CODE	I	Built
FR-OBS-006	Starting-path scoring	Must	CODE	T	In progress
FR-OBS-007	Result screen	Must	CODE	I	In progress
FR-OBS-008	Path never locks activities	Must	RB, README	T	Planned
FR-OBS-009	Change path later	Should	CODE	D	Planned
FR-OBS-010	Save-failure message	Should	CODE	T	In progress
FR-HOM-001	Saga-map home	Must	UI	I	Planned
FR-HOM-002	Fixed stop positions	Must	UI, EXP	T	Planned
FR-HOM-003	Nine stops, no scrolling	Must	UI	D	Planned
FR-HOM-004	Two-level navigation	Must	UI	D	Planned
FR-HOM-005	Start at the suggested path	Must	CODE, UI	T	Planned
FR-HOM-006	Lock map to one stop	Should	UI	D	Planned
FR-HOM-007	Avatar and name on map	Could	LEG	I	Planned

ID	Title	Pri	Source	Method	Status
FR-HOM-008	Scenery only on map	Must	UI	I	Planned
FR-ACT-001	Approved activity sheet	Must	RB	I (E4)	Planned
FR-ACT-002	One instruction, demo, voice	Must	RB, UI	I	Planned
FR-ACT-003	Task fully visible	Must	RB, EXP, LEG	D	Planned
FR-ACT-004	Targets never move	Must	EXP, UI	T	Planned
FR-ACT-005	Tap-tap alternative	Must	RB, EXP, UI	D	Planned
FR-ACT-006	Touch-down, simple gestures	Must	UI	I	In progress
FR-ACT-007	Gentle wrong answers	Must	UI	D	Planned
FR-ACT-008	Feedback within 0.5 s	Must	UI	A	Planned
FR-ACT-009	Pause always available	Must	RB, UI	D	Planned
FR-ACT-010	Resume after interruption	Must	RB	T	Planned
FR-ACT-011	Trial record fields	Must	RB, EXP	T	Planned
FR-ACT-012	One difficulty factor at a time	Should	RB	I	Planned
FR-ACT-013	One quiet mascot	Must	UI	I	Planned
FR-ACT-014	Plain task screens	Must	UI	I	Planned
FR-ACT-015	Vowelised Naskh for learning text	Must	UI	I	Planned
FR-ACT-016	Swap mascot for lead guide	Should	UI	D	Planned
FR-ACT-017	Ink strokes saved and replayed	Must	EXP, RB	T	Planned
FR-ACT-018	Stroke tolerance	Should	RB	T (E4)	Planned

A.2 Activities, rewards, adult area, reports, settings, audio, sync

ID	Title	Pri	Source	Method	Status
FR-EMO-001	Emotions: match the feeling	Must	PLAN, LEG, RB	D	Planned

ID	Title	Pri	Source	Method	Status
FR-EMO-002	Emotions: no colour meaning; records	Must	RB	I	Planned
FR-ENV-001	Environments: place items	Must	PLAN, LEG, RB, EXP	D	Planned
FR-ENV-002	Environments: concept vs missed drop	Must	RB	T	Planned
FR-PTH-001	Path: draw a route	Must	PLAN, LEG, RB	D	Planned
FR-PTH-002	Path: records and wording	Must	RB	I	Planned
FR-STR-001	Star: follow the star	Must	PLAN, LEG, RB	D	Planned
FR-STR-002	Star: records, no attention claim	Must	RB	I	Planned
FR-MAZ-001	Maze: draw, then run the ball	Must	PLAN, LEG, RB	D	Planned
FR-MAZ-002	Maze: drawing vs outcome	Must	RB	T	Planned
FR-GDN-001	Garden: select in order, untimed	Must	PLAN, LEG, RB, EXP	D	Planned
FR-GDN-002	Garden: records	Must	RB	T	Planned
FR-LET-001	Letters: four task types	Must	PLAN, LEG, RB	D	Planned
FR-LET-002	Letters: approved vector forms	Must	EXP, RB	I (E4)	Planned
FR-PAT-001	Patterns: two modes	Must	PLAN, LEG, RB	D	Planned
FR-BAL-001	Balance: equal quantities	Must	PLAN, LEG, RB	D	Planned
FR-RWD-001	Token board	Must	UI	T	Planned
FR-RWD-002	Per-trial celebration	Must	UI	A	Planned
FR-RWD-003	Per-session celebration	Must	UI	D	Planned
FR-RWD-004	Badges with one name each	Should	UI, LEG	I	Planned
FR-RWD-005	Celebration intensity	Must	UI	D	Planned

ID	Title	Pri	Source	Method	Status
FR-RWD-006	Informative praise	Must	UI, RB	I	Planned
FR-RWD-007	No leaderboards or shops	Must	UI, README, RB	I	Built
FR-RWD-008	Own-progress stars	Must	UI	T	Planned
FR-ADM-001	Adult area behind gate	Must	UI	T	Planned
FR-ADM-002	Gate resists random input	Must	LEG, UI	T	Planned
FR-ADM-003	Gate is not consent	Must	RB	I	Planned
FR-ADM-004	Child can stop without gate	Must	RB, UI	D	Planned
FR-ADM-005	Accessible gate	Should	RB	D	Planned
FR-ADM-006	One action back to child mode	Should	LEG	D	Planned
FR-REP-001	Counts beside percentages	Must	README, RB	I	Planned
FR-REP-002	No unexplained composites	Must	RB, LEG	I	Planned
FR-REP-003	No ranking of children	Must	README, RB	I	Planned
FR-REP-004	Skill named, no diagnosis	Must	RB	I	Planned
FR-REP-005	Session list	Should	LEG, RB	T	Planned
FR-REP-006	Like-for-like trends	Should	LEG, RB	T	Planned
FR-REP-007	Copy summary	Could	LEG	T	Planned
FR-REP-008	Report read correctly by adults	Should	RB	D (E3)	Planned
FR-SET-001	Three comfort modes	Must	UI, CODE	A	In progress
FR-SET-002	Per-child settings	Must	UI	T	In progress
FR-SET-003	Settings persist in order	Must	CODE	T	Built
FR-SET-004	Three motion levels	Must	UI	T	In progress
FR-SET-005	Sensory starting point	Should	UI	D	Planned

ID	Title	Pri	Source	Method	Status
FR-SET-006	Music off, voice on by default	Must	UI, CODE	T	Built
FR-AUD-001	Three audio channels	Must	UI, RB, CODE	T	In progress
FR-AUD-002	Soft effects, speech on top	Must	UI	A (E2)	In progress
FR-AUD-003	Recorded, approved voice	Must	README, RB, CODE	I (E4)	In progress
FR-AUD-004	Usable with audio off	Must	RB	D	In progress
FR-AUD-005	Audio works offline	Must	RB	T	In progress
FR-AUD-006	Soft pop on press	Should	UI, CODE	D	Built
FR-SYN-001	Child mode offline	Must	RB	D	In progress
FR-SYN-002	Write locally first	Must	EXP	T	Planned
FR-SYN-003	Idempotent upload	Must	EXP	T	Planned
FR-SYN-004	Consent before sync	Must	RB, README	I	Planned
FR-SYN-005	Multi-device conflict rule	Should	EXP, LEG	T	Planned

A.3 Interfaces and non-functional requirements

ID	Title	Pri	Source	Method	Status
IR-UI-001	Always RTL	Must	README, CODE	T	Built
IR-UI-002	Rounded heavy interface face	Must	UI, CODE	I	Built
IR-UI-003	Minimum label sizes	Must	UI	I	Built
IR-UI-004	Corners and depth	Must	UI, CODE	A	In progress
IR-UI-005	Chunky button behaviour	Must	UI, CODE	D	Built
IR-UI-006	Lip edge on light faces	Must	UI, WCAG, CODE	A	Built

ID	Title	Pri	Source	Method	Status
IR-UI-007	One hue per activity	Must	UI	I	In progress
IR-UI-008	What mirrors in RTL	Must	UI, CODE	I	Built
IR-UI-009	Edge insets	Must	UI	A	Built
IR-UI-010	Vector art in code	Must	UI, CODE	I	Built
IR-UI-011	Inactive button style	Should	CODE	I	Built
IR-HW-001	Stylus and finger, tool recorded	Must	EXP	D (E2)	Planned
IR-HW-002	Palm and stylus-only tested	Must	EXP, RB	D (E2)	Planned
IR-HW-003	Latency threshold agreed	Must	RB	A (E2)	Planned
IR-HW-004	Left- and right-handed reach	Should	RB	D (E2)	Planned
IR-HW-005	Pressure and tilt only if reliable	Could	RB	A (E2)	Planned
IR-AUD-001	System audio streams and volume	Must	CODE	T	Built
IR-AUD-002	Silent without Arabic voice	Must	CODE	T	In progress
IR-STO-001	Local Room database	Must	CODE	I	Built
IR-STO-002	Migrations keep data	Must	CODE, EXP	T	Built
IR-STO-003	Trials keyed by child and ID	Must	EXP, RB	I	Planned
IR-STO-004	Private storage, no backup	Must	RB, CODE	I	Built
NFR-ACC-001	WCAG contrast in all modes	Must	WCAG, UI, CODE	A	Built
NFR-ACC-002	Colour never the only cue	Must	WCAG, UI	I	Built
NFR-ACC-003	Target sizes and gaps	Must	UI, EXP, RB	A	In progress
NFR-ACC-004	No scaled text in target areas	Must	EXP	I	Planned
NFR-ACC-005	Works at font scale 1.5	Must	RB, EXP	D	In progress
NFR-ACC-006	Reduced motion, no flashing	Must	UI, WCAG	D	Built

ID	Title	Pri	Source	Method	Status
NFR-ACC-007	Purposeful, short motion	Must	UI	A	Planned
NFR-ACC-008	One accessible node per element	Must	EXP, RB	I	In progress
NFR-ACC-009	TalkBack tested on device	Must	RB, EXP	D (E2)	Planned
NFR-ACC-010	Visible focus	Should	RB, EXP	D (E2)	Planned
NFR-PRF-001	Ink keeps up	Must	EXP, RB	A (E2)	Planned
NFR-PRF-002	Non-blocking, ordered writes	Must	CODE, RB	T	Built
NFR-OFF-001	Everything bundled	Must	RB, LEG	D	In progress
NFR-OFF-002	No network permission yet	Must	RB, CODE	I	Built
NFR-PRV-001	Data inventory	Must	RB	I (E4)	Planned
NFR-PRV-002	No AI or analytics	Must	RB, LEG	I	Built
NFR-PRV-003	Answers treated as sensitive	Must	RB	I	In progress
NFR-PRV-004	Separate gate, research, cloud consent	Must	RB	I	Planned
NFR-PRV-005	Deletion and retention policy	Must	RB	I (E4)	Planned
NFR-PRV-006	No child data in logs	Must	RB	T	Built
NFR-PRV-007	PDPL review	Must	RB	I (E4)	Planned
NFR-PRV-008	Field-session safeguards	Must	RB	I (E3)	Planned
NFR-SAF-001	No clinical claims	Must	README, RB	I	In progress
NFR-SAF-002	No derived diagnoses or ages	Must	RB	I	Built
NFR-SAF-003	Stop without penalty	Must	RB	D	Planned

ID	Title	Pri	Source	Method	Status
NFR-LOC-001	Arabic only	Must	README	I	Built
NFR-LOC-002	One numeral system	Must	UI, RB	I	In progress
NFR-LOC-003	Tashkeel checked	Must	UI, RB	I (E4)	Planned
NFR-LOC-004	Correct mixed-direction text	Must	RB, CODE	D	Built
NFR-MNT-001	Self-contained `android/` project	Must	README	I	Built
NFR-MNT-002	Tokens in one file, checked	Must	CODE	I	Built
NFR-MNT-003	Unit tests	Should	ISO, CODE	T	Planned
NFR-MNT-004	Strings in resources	Should	README	I	Planned
NFR-MNT-005	Pinned versions, Ink recheck	Must	EXP	I	In progress
NFR-MNT-006	Onboarding docs current	Should	CODE	I	Planned

Appendix B. Open issues and pending decisions

Twenty decisions are left open because the sources conflict or are silent; requirements that depend on them cite the issue ID and must not be closed until it is decided.

ID	Issue	What the sources say	Affects	Needed from
OI-01	Age range	Plan: 4–10 [PLAN §Limits]. Legacy form: 2–6 [LEG §4]. Code follows the legacy 2–6. Research: confirm through curriculum and participants, not the old slider [RB §3].	FR-PRF-002, target sizes, content	Owner, educators

ID	Issue	What the sources say	Affects	Needed from
OI-02	Numeral system	Western or Arabic-Indic is unresolved [README; RB §6; UI §Carried forward]. Code shows Western digits.	NFR-LOC-002, FR-PRF-002, maths activities	Owner, Arabic educator
OI-03	Guide name	Code uses «نُهى», the product name, as a working name. No source names the guide.	FR-ONB-003, voice script	Owner
OI-04	Voice register and recordings	Fusha or dialect is open [UI §Open decisions]. The TTS stand-in picks an Egyptian-dialect voice (`ar-xa-x-arz`) on the emulator.	FR-AUD-003, all prompts	Owner, Arabic educator
OI-05	Observation questions and scoring	Questions reworded from the legacy form so «نعم» means more difficulty; area mapping and thresholds (0.5 lead, 0.15 margin) are placeholders. Legacy algorithm unknown [LEG §4].	FR-OBS-002, FR-OBS-006	Specialist, Arabic educator
OI-06	Stylus-only or finger allowed	Plan is stylus-based [PLAN]; README and research ask whether touch is an allowed alternative [README; RB §3, §10].	IR-HW-001/002, FR-ACT-005	Owner

ID	Issue	What the sources say	Affects	Needed from
OI-07	Table hardware	No device or pen selected; table build status unknown [README; RB §10; LEG §1].	All E2 checks, layout geometry	Owner, hardware reviewer
OI-08	Operator	Family, institution or service provider; changes account ownership and data responsibility [README; RB §12].	FR-PRF-008, NFR-PRV-*, sync	Owner
OI-09	Curriculum and content reviewer	Saudi curriculum, register and reviewer not named [README; RB §6].	FR-ACT-001, FR-LET-*, NFR-LOC-003	Owner
OI-10	Final activity list and objectives	Nine activities named; each needs one objective [README; RB §8]. Maze mechanic conflicts: walls [PLAN] vs a drawn route [LEG §6.6]. Legacy extras «فراشة التركيز» and «تحدّي التركيز» not approved [README].	§3.1.5	Owner, educators
OI-11	Interface font	Baloo Bhaijaan 2 vs Readex Pro after a harakat collision test on device [UI §Open decisions]. Code uses Baloo Bhaijaan 2.	IR-UI-002	Owner, Arabic educator
OI-12	Mascot cast	One mascot per activity or fewer shared ones [UI §Open decisions].	FR-ACT-013/016, art budget	Owner

ID	Issue	What the sources say	Affects	Needed from
OI-13	Map art budget	One landscape or one per activity group [UI §Open decisions].	FR-HOM-001	Owner
OI-14	Motion default and sensory starting point	Style doc: motion defaults to the system value, and a therapist picks lively, gentle or muted before the first session [UI §Motion, §Conflicts]. Code: reduced motion on by default; no starting-point step in the child-first onboarding [CODE; OWN].	FR-SET-004/005, FR-ONB-*	Owner
OI-15	Gate mechanism	Legacy uses an addition question [LEG §5]; a 3-second hold was built and removed [OWN]; a gate is not guardianship verification [RB §12].	FR-ADM-002/005	Owner
OI-16	Accounts and sign-in	Legacy requires a parent sign-in [LEG §3]; the new app is local-first with no account.	Adult area, sync, consent	Owner, PDPL reviewer
OI-17	Sync backend and conflict rule	Sync to be added only once benefit and access model are agreed [RB §11]; multi-device conflicts untested [EXP §4].	FR-SYN-003–005	Owner

ID	Issue	What the sources say	Affects	Needed from
OI-18	"Therapeutic" wording	The plan calls the games "educational and therapeutic" [PLAN §Abstract]; README and research rule out treatment claims [README; RB §1].	NFR-SAF-001, store copy	Owner
OI-19	Session length and breaks	Legacy plans three short activities with breaks [LEG §5]; research finds no justified session duration [RB §10].	Home flow, rewards break	Owner, educators
OI-20	Activities per starting path	Plan groups activities by diagnosis [PLAN]; legacy by skill programme, with four communication games [LEG §5]; research rejects diagnosis-based gating [RB §7]. The mapping of activities to the three paths is undefined.	FR-HOM-005, FR-OBS-008	Owner, educators